DGMS 2019 GEOLOGY • QUESTION 34

The Earth's Lithosphere: Mechanical vs. Compositional Layering

Topic: Geophysics, Petrology & Geodynamics | Primary Reference: John D. Winter (Principles of Igneous & Metamorphic Petrology, Ch. 1)

Question Statement

Q.34. The lithosphere is comprised of

(a) Mantle and outer core
(b) Mantle and asthenosphere
(c) Upper mantle and crust
(d) Continental and oceanic crusts

a Mantle and outer core	b Mantle and asthenosphere
c **Upper mantle and crust (Correct Answer)**	d Continental and oceanic crusts

✓ Correct Answer: (c) Upper mantle and crust [Specifically the crust plus the uppermost rigid lithospheric mantle above the asthenosphere].

Textbook Evidence: Winter (Chapter 1)

gradually by plate tectonic processes: oceanic crust at divergent plate boundaries and continental crust as small island-arc terranes at convergent (subduction) boundaries, which accreted by collisions to form larger continental masses.

of the overriding plate at subduction zones)
- Hotspots (rising plumes that can penetrate either oceanic or continental lithospheric plates)

Key Terms

Magma
Lava
Volcanic
Plutonic
Oceanic crust
Continental crust
Shield
Platform
Mantle
Core
Lithosphere
Asthenosphere
Low-velocity layer
Lithophile
Chalcophile
Siderophile
Iron meteorites
Stone meteorites
Stony-iron meteorites
Chondrite meteorites
Lithostatic pressure
Conduction of heat
Convection of heat
Radiation of heat
Advection of heat
Geothermal gradient
Slab-pull/ridge-push
Petrogenesis
Igneous-tectonic association

Review Questions and Problems

Review Questions and Problems are located on the author's web page at the following address: **http://www.prenhall.com/winter**

Important "First Principle" Concepts

- Igneous rocks crystallize from a melt. They can be recognized either by field setting or by textural criteria.
- The oceanic crust, continental crust, mantle, and core are the four principle *compositional* subdivisions of the Earth.
- The lithosphere, asthenosphere, mesosphere, outer core, and inner core are the five principal *mechanical* subdivisions of the Earth.

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Figure 1-11: Rheological and Compositional Subdivisions of the Earth (John D. Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed., p. 29). Cross-section illustrating the fundamental duality of the Earth's upper layers: compositionally, the boundary between the crust and mantle is the *Mohorovičić discontinuity (Moho)*; mechanically, the rigid **lithosphere** consists of the entire crust plus the uppermost rigid mantle (the lithospheric mantle), resting above the ductile, convecting **asthenosphere**.

> *"The* **lithosphere** *(from Greek lithos, stone) is the rigid, mechanically strong outer layer of the Earth. It includes the* **crust** *and the* **uppermost solid, rigid portion of the upper mantle***. The underlying* **asthenosphere** *(from Greek asthenes, weak) is part of the upper mantle, but differs rheologically: it is plastic, ductile, and capable of flow over geologic timescales."*
>
> **— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 1, pp. 11–12**

Comprehensive Option Analysis

(c) Upper mantle and crust **[CORRECT]**

Rheological & Structural Architecture: Earth is layered in two distinct ways:

- **Compositional (Chemical) Layering:** Based on density and mineral chemistry—**Crust** (felsic to mafic silicates), **Mantle** (ultramafic peridotite), and **Core** (iron-nickel metal).
- **Mechanical (Rheological) Layering:** Based on mechanical strength and deformation response—**Lithosphere** (rigid/brittle plate), **Asthenosphere** (weak/ductile mantle), **Mesosphere** (stiff lower mantle), **Outer core** (liquid), and **Inner core** (solid).

The **lithosphere** is defined precisely as the rigid outer shell that moves as coherent tectonic plates. It encompasses the entirety of the **crust** (both continental and oceanic) plus the **uppermost mantle** (lithospheric mantle / LID), bounded below by the ~1300°C isotherm that marks the transition into the ductile asthenosphere.

(a) Mantle and outer core **[INCORRECT]**

The outer core is a liquid iron-nickel metallic alloy extending from 2,890 km to 5,150 km depth. The lithosphere is confined exclusively to the outermost 100–250 km of the Earth.

(b) Mantle and asthenosphere **[INCORRECT]**

The asthenosphere is the *ductile substrate* beneath the lithosphere, not a constituent of it. Moreover, the lithosphere contains the crust, which is omitted here, while the lithosphere does NOT include the entire mantle.

(d) Continental and oceanic crusts **[INCORRECT]**

Continental and oceanic crusts together make up only the **Crust**. Confining the lithosphere to the crust is a classic student pitfall: the crust is merely the top 7–70 km of the lithosphere. The lithosphere extends down to 100 km (beneath oceans) and up to 200–250 km (beneath ancient continental cratons) because it includes the rigid upper mantle.

Compositional vs. Mechanical Layers of the Solid Earth

Classification	Layer Name	Typical Depth	Composition & Physical State
Compositional (Chemical)	Crust (Continental / Oceanic)	0 – 35 km (avg)	Granitic to basaltic silicates; rigid, low density (2.7–3.0 g/cm³)
	Mantle (Upper + Lower)	35 – 2,890 km	Peridotite (olivine, pyroxene, garnet); bridgmanite, ferropericlase (3.3–5.5 g/cm³)
	Core (Outer + Inner)	2,890 – 6,371 km	Iron-nickel alloy + light elements (S, Si, O) (10–13 g/cm³)
Mechanical (Rheological)	**Lithosphere**	**0 – 100/250 km**	**Rigid, elastic, brittle plates: Crust + Uppermost Mantle (LID)**
	Asthenosphere	100 – 410/660 km	Ductile, plastic peridotite; capable of slow solid-state convective flow
	Mesosphere	660 – 2,890 km	Rigid, solid lower mantle under high lithostatic pressure
	Outer Core / Inner Core	2,890 – 6,371 km	Liquid outer core (magnetic dynamo) / Solid inner core

Key Takeaway & Exam Strategy

Core Golden Rule:

- **Moho:** The compositional boundary separating crust from mantle.
- **LAB (Lithosphere-Asthenosphere Boundary):** The rheological/thermal boundary separating the rigid lithosphere (crust + upper mantle) from the ductile asthenosphere.
- Always remember: **Lithosphere = Crust + Uppermost Mantle**.